Hossain Mohammad Jayed

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Profile

CSE undergraduate at Daffodil International University with a growing research track record in applied machine learning. Currently running two studies toward journal publication: one on few-shot rice leaf disease classification, the other on detecting rumors in social media text. Also builds independent Python tools on the side to stay sharp outside coursework.

Research Experience

Undergraduate Researcher – BanglaFSL-Rice *DIU, CSE Dept.*

- Working on ”Attention-Enhanced Prototypical Networks for Explainable Few-Shot Rice Leaf Disease Classification,” aimed at *Computers & Electronics in Agriculture* (Q1, Scopus), with co-researcher Afrin Akter Bresti under faculty supervision on the approved FYDP.
- Built a CBAM-enhanced Prototypical Network on a ResNet-12 + GroupNorm backbone and tested it against Plain ProtoNet, FEAT, RelationNet, and MAML in-domain, across species, and on real field images.
- Traced a stubborn non-determinism bug back to domain shift, then fixed it with seed-controlled Wilcoxon testing, aspect-preserving preprocessing, and Grad-CAM/SHAP checks. Nothing gets written up until it holds across five seeds.

Undergraduate Researcher – Rumor Veracity Detection *DIU, CSE Dept.*

- Co-authoring ”Transformer-Based Rumor Veracity Detection on Social Media: A Study Using PHEME and RumourEval Datasets,” with co-authors Rifat Parvej Pranto and MD Amirul Islam.
- Fine-tuned and compared BERT, RoBERTa, and DeBERTa-v3, using class-weighted and Focal Loss to handle a heavy minority-class imbalance.
- Set up event-grouped data splits, multi-seed evaluation, a soft-voting ensemble, bootstrap confidence intervals, and McNemar’s test to keep the comparisons honest.

Experience & Leadership

Student Associate *HR Department, DIU (3 months)*

- Handled sensitive university records with a focus on accurate, careful data entry.

Executive Member *DIU Computer Programming Club (CPC)*

- Helped organize and run campus tech events, including UTA, Takeoff, and Prompt Battle.

Projects

HAMOZA AI – Telegram Bot Inline-keyboard Telegram bot on `pyTelegramBotAPI`; token loaded from environment variables, not hardcoded.

Jarvis – Voice-Controlled Desktop Assistant Voice assistant for Wikipedia lookups, browsing, and email, built with `speech_recognition` and `pyttsx3`.

Phone Number Intelligence Tool Resolves carrier, timezone, and location from a phone number via `phonenumbers` and the OpenCage API, plotted on a `folium` map.

Password Strength Analyzer Brute-force keyspace search benchmarking crack time against password length and complexity.

Additional Utilities PDF merger, desktop notifier, language detector, translator, Roman-numeral converter, BMI calculator, and small games.

Certifications

CS50P: Introduction to Programming with Python (HarvardX) | AWS Academy Cloud Foundations | Introduction to Modern AI (Wadhwani Foundation) | Introduction to Data Science (Wadhwani Foundation)

Skills

Languages: Python (Advanced), C, SQL, JavaScript (ES6+)
ML / NLP: PyTorch, Few-Shot Learning (ProtoNet, MAML), Transformer fine-tuning (BERT/RoBERTa/DeBERTa-v3), Grad-CAM/SHAP, scikit-learn
Tools: Git, GitHub, Jupyter/Kaggle, MS Office
Core CS: DSA, Databases, OS, Computer Networks, Software Engineering
Spoken Languages: Bangla (Fluent), English (Working Proficiency), Hindi (Basic)

Education

Daffodil International University 2023 – 2026 (Expected)
B.Sc. in Computer Science and Engineering, Section 64_A

Cantonment College, Jashore – HSC in Science 2019 – 2021

Akij Collegiate School – SSC in Science 2017 – 2019